

# LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

## Problem Statement #22 – Smart Parking Space Rental & License Validator

### Architecture Selection

We chose **Microservices Architecture** for the Smart Parking Space Rental & License Validator.

### Architectural Choice

The system is divided into separate services for reservation, license-plate validation, and parking monitoring and fee processing. Vehicle Owner and Warden interfaces communicate with these services through defined interfaces, while the Parking Database stores vehicle, reservation, entry/exit, fee and overstay information.

### Reason 1 — Real-Time License Validation

License-plate validation is a critical gate operation. Keeping validation as a separate service allows this function to be handled independently and supports the requirement that 95 out of 100 verification requests complete within 300 ms under normal conditions.

### Reason 2 — Independent Service Handling

Reservation, license validation, and parking monitoring have different responsibilities. Separating them into services allows each function to be maintained and scaled independently when the workload changes.

### Security Advantage

Access to vehicle, reservation and parking information can be controlled through the service interfaces. This supports the requirement that only authorized users can access protected parking information.

### Performance Benefit

The License Plate Validation Service can process gate-camera requests without requiring the complete reservation and monitoring workflow to execute at the gate. This supports fast gate processing and helps meet the 300 ms verification requirement.